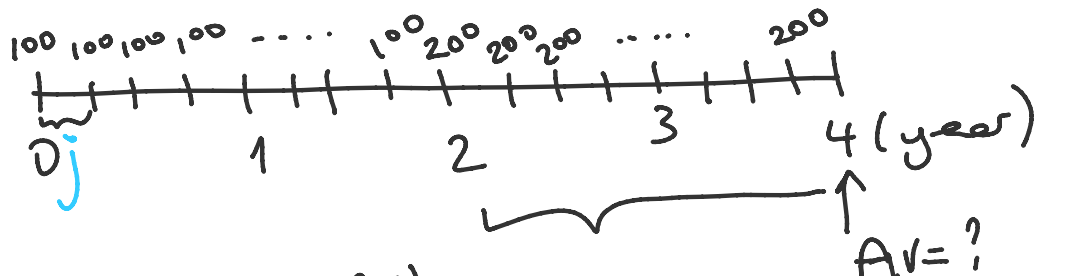


$$A.V = ?$$

Example 4.1 Find the accumulated value at the end of four years of an investment fund in which \$100 is deposited at the beginning of each quarter for the first two years and \$200 is deposited at the beginning of each quarter for the second two years, if the fund earns 12% convertible monthly.

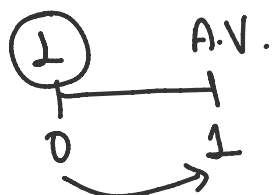


$$m=4$$

$$i^{(12)} = 12\%$$

$$i^{(4)} = ?$$

$$\frac{i^{(4)}}{4} = j$$



$$AV = \left(1 + \frac{i^{(4)}}{4}\right)^4 = \left(1 + \frac{i^{(12)}}{12}\right)^{12} = AV \text{ where } i^{(12)} = 12\%$$

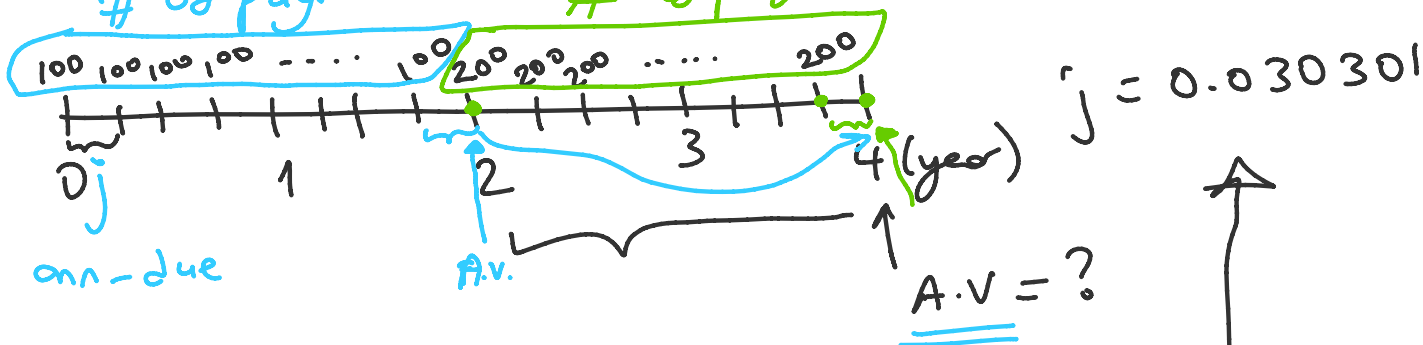
$$(1+j)^4 = \left(1 + \frac{0.12}{12}\right)^{12}$$

Solve this equation for j .

$$j = (1.01)^3 - 1$$

$$j = (1.01)^3 - 1 = .030301.$$

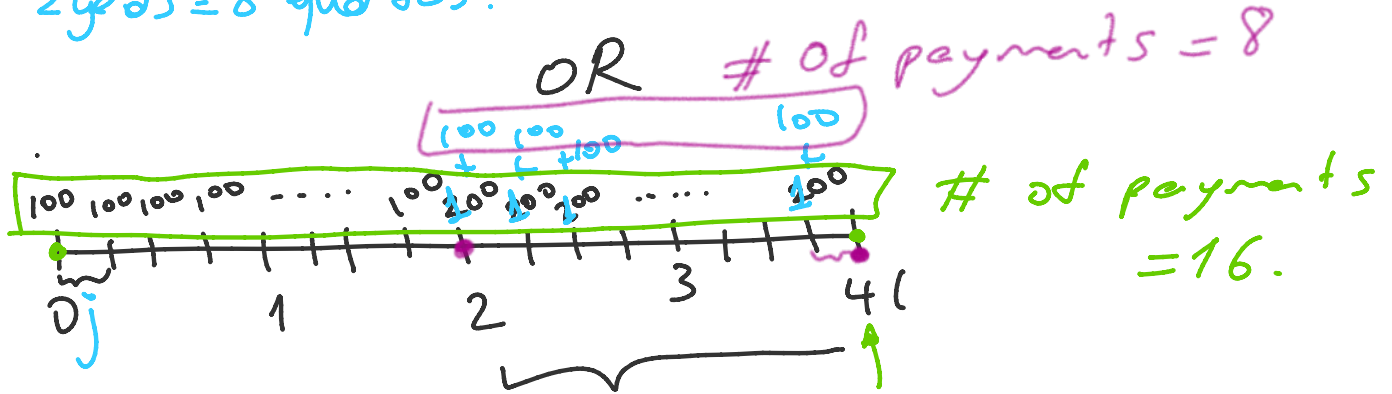
of payments = 8 # of payments = 2 x 4 = 8.



$$100 \cdot \ddot{s}_{\overline{8}|j} \cdot (1+j)^8 + 200 \cdot \ddot{s}_{\overline{8}|j} = A.V. \text{ where}$$

$$100 \cdot \ddot{s}_{\overline{8}|j} \cdot \underbrace{(1+j)}_{\substack{\text{per} \\ \text{1-quarter}}} + 200 \cdot \ddot{s}_{\overline{8}|j} = A.V. \text{ where}$$

2 years = 8 quarters.



$$100 \cdot \ddot{s}_{\overline{16}|j} + 100 \cdot \ddot{s}_{\overline{8}|j} = A.V.$$

We are given an interest rate of 1% per month. Let j be the equivalent rate of interest per quarter, which is the payment period. We have

$$j = (1.01)^3 - 1 = .030301.$$

The value of the annuity in symbols is

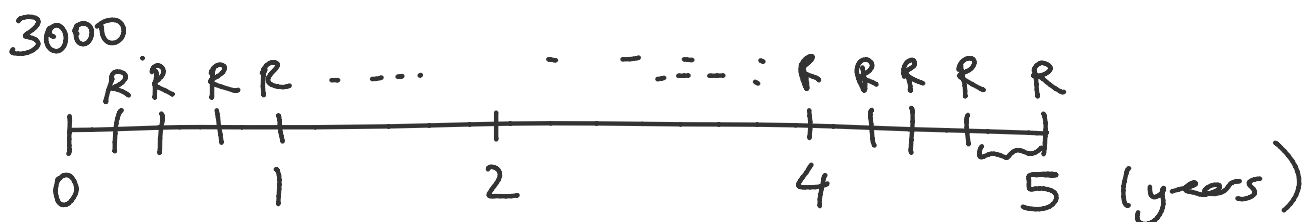
$$100 \left(\ddot{s}_{\overline{16}|j} + \ddot{s}_{\overline{8}|j} \right)$$

which can be evaluated as

$$100(20.81704 + 9.17157) = \$2998.86.$$

3

Example 4.2 A loan of \$3000 is to be repaid with quarterly installments at the end of each quarter for five years. If the rate of interest charged on the loan is 10% convertible semiannually, find the amount of each quarterly payment.



$$\frac{j^{(4)}}{4} = i = ?$$

$$i^{(2)} = 10\%.$$

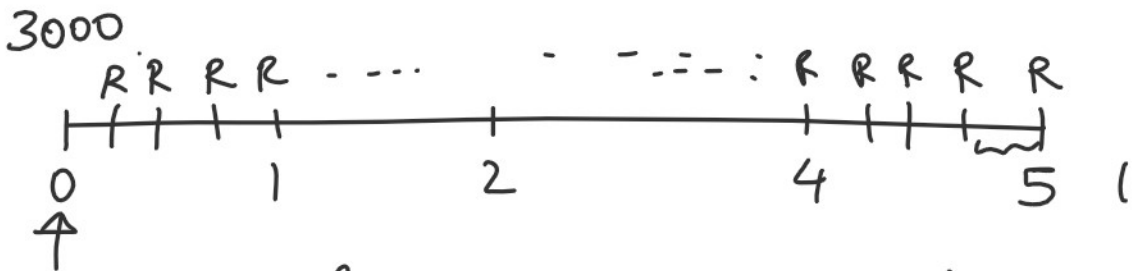
$$\frac{i^{(4)}}{4} = j = ?$$

$$i^{(2)} = 10\%$$

$$\left(1 + \frac{i^{(4)}}{4}\right)^4 = \left(1 + \frac{i^{(2)}}{2}\right)^2 \text{ where } i^{(2)} = 10\%$$

$$j = (1.05)^{\frac{1}{2}} - 1$$

$$j = (1.05)^{1/2} - 1 = .024695.$$



$$\# \text{ of payments} = 5 \times 4 = 20.$$

$$R \cdot a_{\overline{20}|j} = 3000$$

We are given an interest rate of 5% per half-year. Let j be the equivalent rate of interest per quarter, which is the payment period. We have

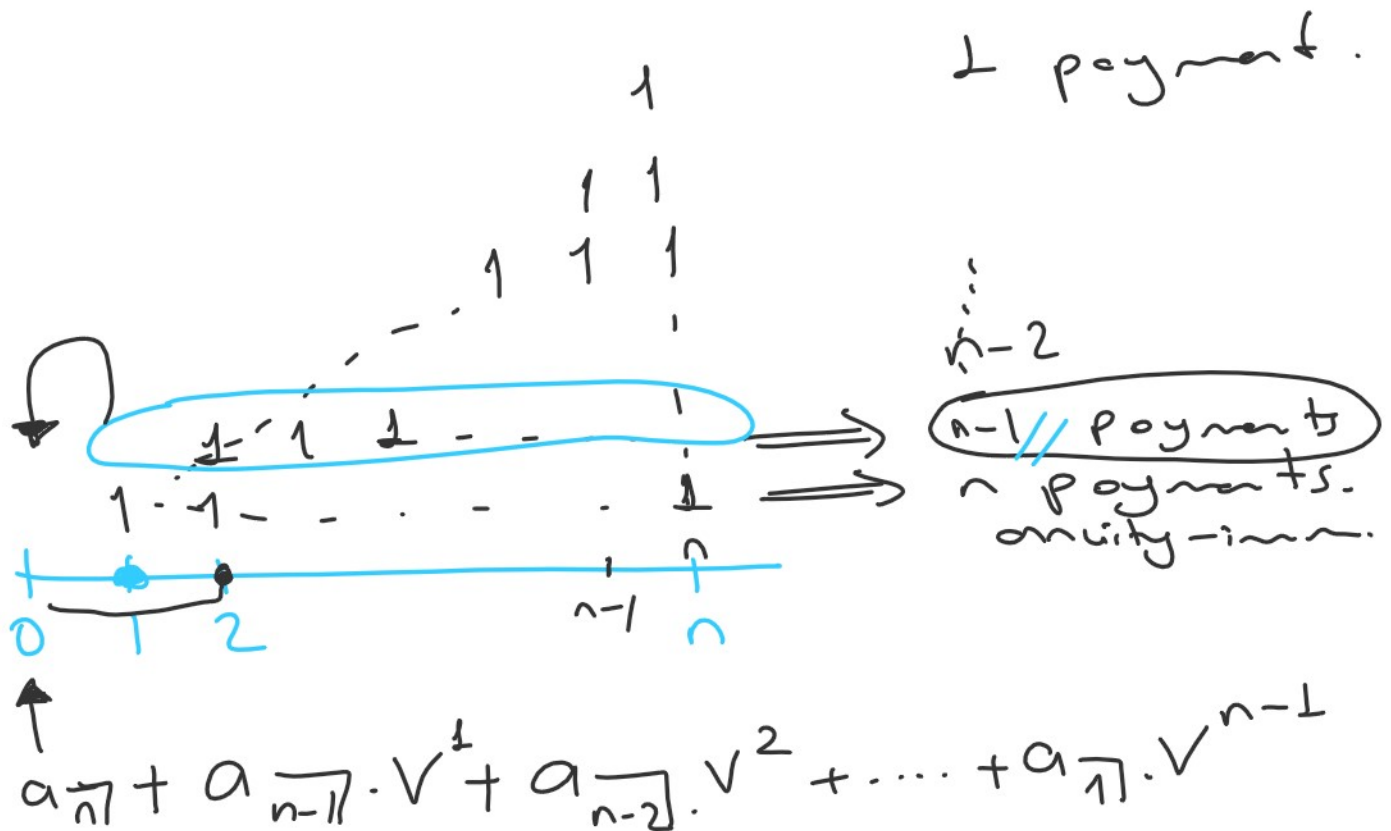
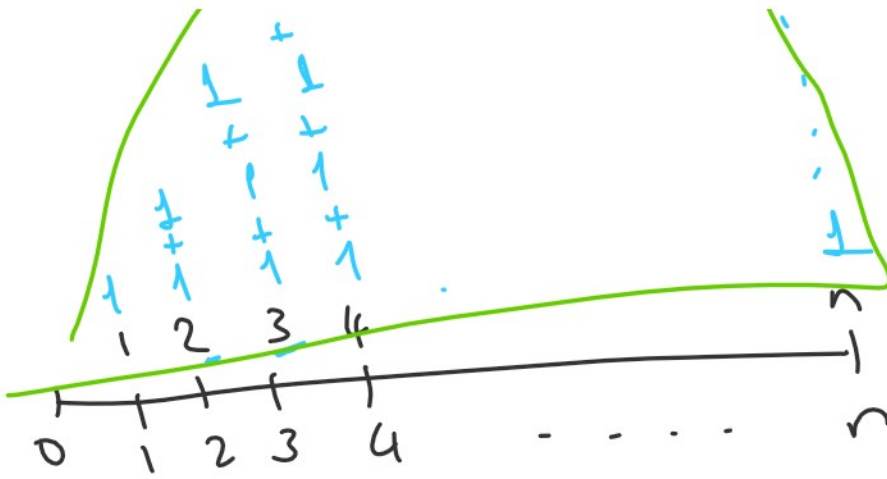
$$j = (1.05)^{1/2} - 1 = .024695.$$

Let the quarterly payment be denoted by R . Then the equation of value is

$$Ra_{\overline{20}|j} = 3000$$

so that

$$R = \frac{3000}{a_{\overline{20}|j}} = \frac{3000}{15.63417} = \$191.89.$$



$$A = Pa_{\overline{n}|} + Q \frac{a_{\overline{n}|} - nv^n}{i}$$

$$n \rightarrow \infty. \quad \lim. \quad a_{\overline{\infty}|} = \frac{1}{i}$$

$$\lim_{n \rightarrow \infty} \left(P \cdot a_{\overline{n}|} + Q \cdot \frac{a_{\overline{n}|} - nv^n}{i} \right)$$

$$n \rightarrow \infty \quad \left(\frac{1}{i} \right)$$

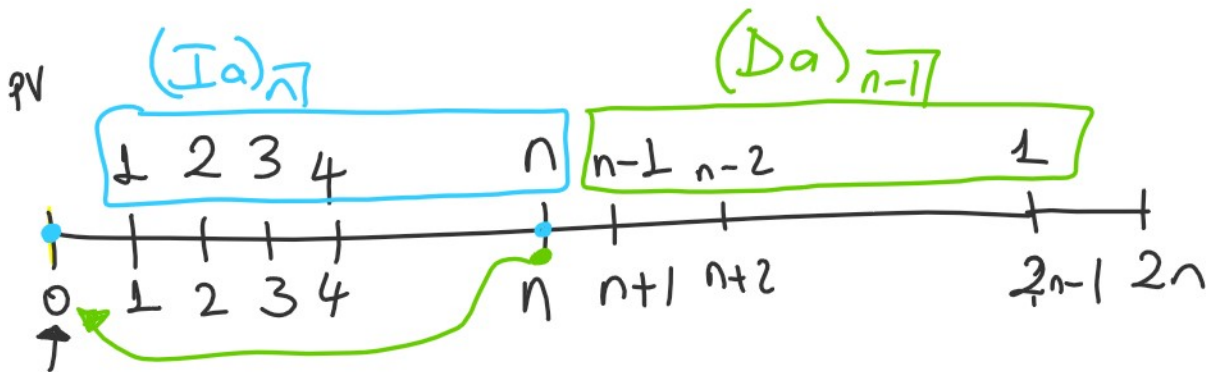
$$= \frac{P}{i} + \frac{Q}{i^2}$$

$$v = \frac{1}{1+i} < 1$$

$$\lim_{n \rightarrow \infty} n \cdot v^n = \lim_{n \rightarrow \infty} \frac{v^n}{\frac{1}{n}} \rightarrow \frac{0}{0} \quad \text{L'Hop.}$$

$$= 0.$$

Example 4.10 Find the present value of an annuity-immediate such that payments start at 1, increase by annual amounts of 1 to a payment of n , and then decrease by annual amounts of 1 to a final payment of 1.



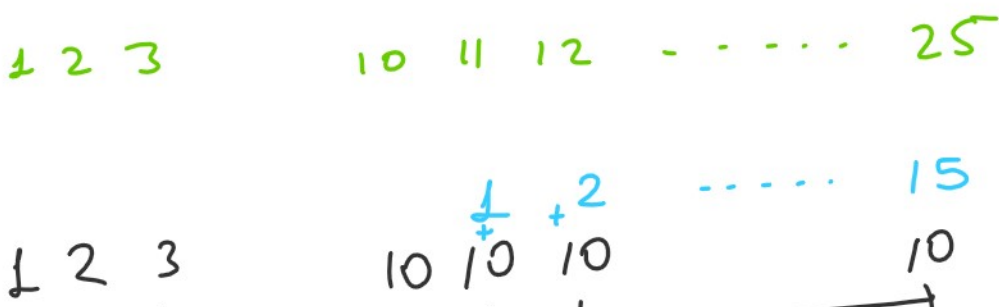
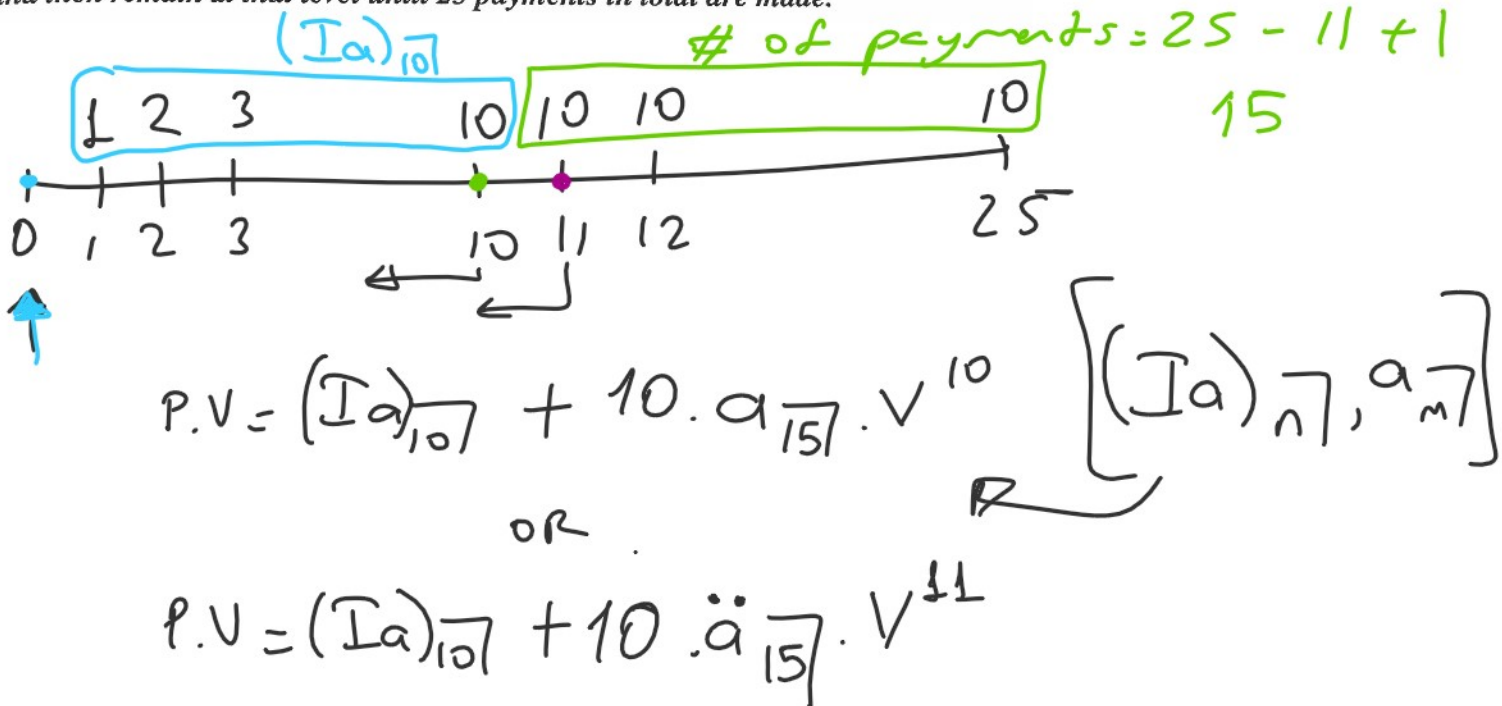
$$(Ia)_n + (Da)_{n-1} \cdot v^n$$

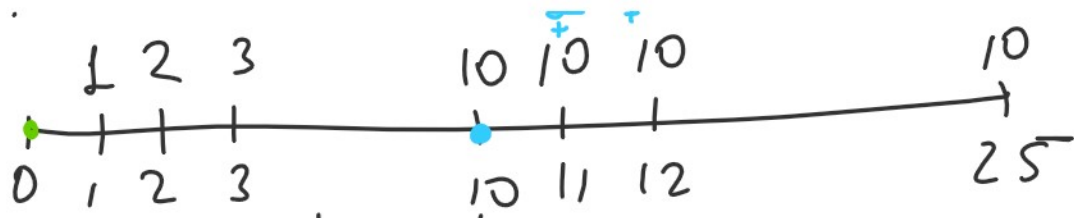
Example 4.10 Find the present value of an annuity-immediate such that payments start at 1, increase by annual amounts of 1 to a payment of n , and then decrease by annual amounts of 1 to a final payment of 1.

The present value is

$$\begin{aligned}
 (Ia)_{\overline{n}|} + v^n (Da)_{\overline{n-1}|} &= \frac{\ddot{a}_{\overline{n}|} - nv^n}{i} + v^n \frac{(n-1) - a_{\overline{n-1}|}}{i} \\
 &= \frac{1}{i} [a_{\overline{n-1}|} + 1 - nv^n + nv^n - v^n - v^n a_{\overline{n-1}|}] \\
 &= \frac{1}{i} [a_{\overline{n-1}|} (1 - v^n) + (1 - v^n)] \\
 &= \frac{1}{i} (1 - v^n) (a_{\overline{n-1}|} + 1) \\
 &= a_{\overline{n}|} \cdot \ddot{a}_{\overline{n}|}.
 \end{aligned}$$

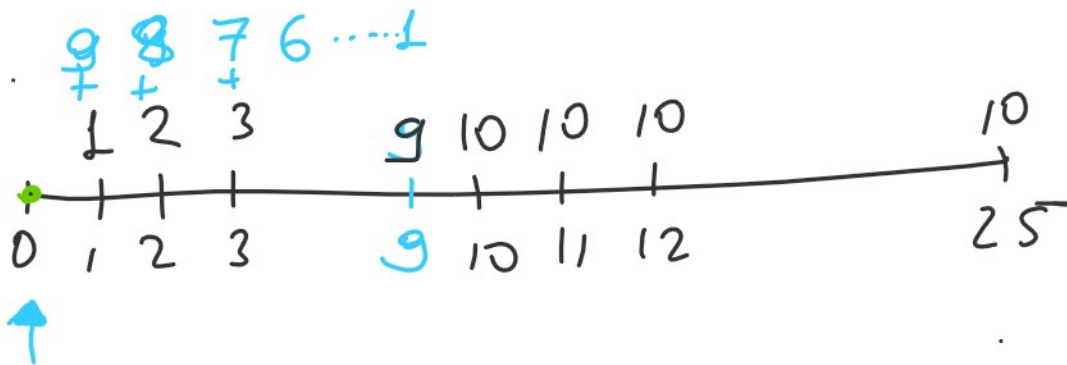
Example 4.11 Find the present value of an annuity-immediate such that payments start at 1, each payment thereafter increases by 1 until reaching 10, and then remain at that level until 25 payments in total are made.





$$P.V = (Ia)_{\overline{25}|} - (Ia)_{\overline{15}|} \cdot V^{10}$$

10 10 10 10 - - - - - 10) # 25



$$PV = 10 \cdot a_{\overline{25}|} - (Da)_{\overline{9}|}$$

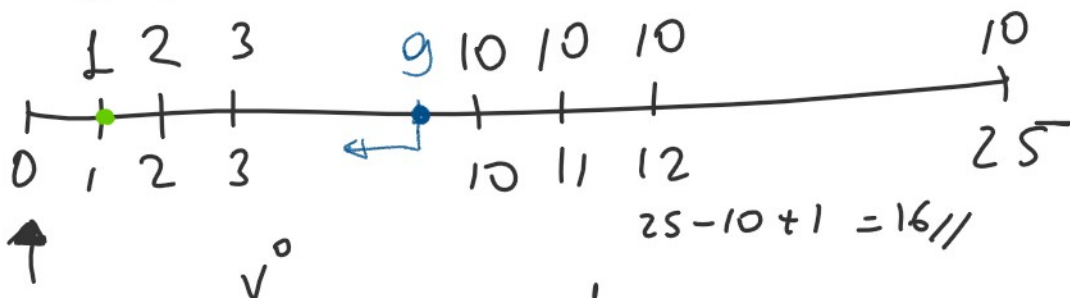


1 1 1 - - - - - 1 $\Rightarrow 16$

1 - - - - - 1 $\Rightarrow$

1 1 1 - - - - - 1 $\Rightarrow 24$

1 1 1 - - - - - 1 $\Rightarrow 25$



$$PV = a_{\overline{25}|} + a_{\overline{24}|} \cdot V^1 + a_{\overline{23}|} \cdot V^2 + \dots + a_{\overline{16}|} \cdot V^9$$

$$- \frac{9}{V} V^t a_{\overline{16}|}$$

$$= \sum_{t=0}^9 v^t a_{\overline{25-t}|}$$

Example 4.11 Find the present value of an annuity-immediate such that payments start at 1, each payment thereafter increases by 1 until reaching 10, and then remain at that level until 25 payments in total are made.

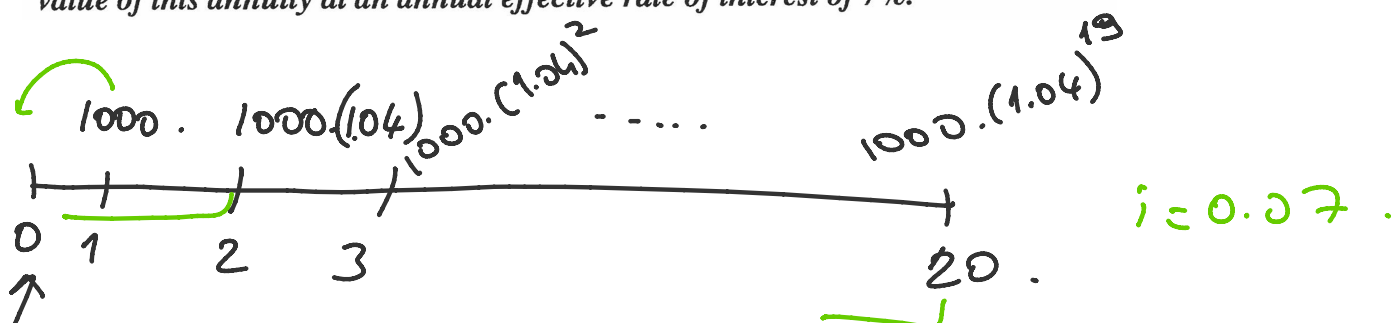
A symbol for an n -period increasing annuity-immediate in which increases are limited to the first m periods only, $0 < m < n$, is given by

$$(I_{\overline{m}|}a)_{\overline{n}|}$$

Thus, the present value of the annuity in this example can be written as $(I_{\overline{10}|}a)_{\overline{25}|}$. The reader should verify that the following are all valid expressions for the present value of this annuity:

- $(Ia)_{\overline{10}|} + 10v^{10}a_{\overline{15}|}$
- $(Ia)_{\overline{25}|} - v^{10}(Ia)_{\overline{15}|}$
- $10a_{\overline{25}|} - (Da)_{\overline{9}|}$
- $\sum_{t=0}^9 v^t a_{\overline{25-t}|}$

Example 4.12 An annuity provides for 20 annual payments, the first payment a year hence being \$1000. The payments increase in such a way that each payment is 4% greater than the preceding payment. Find the present value of this annuity at an annual effective rate of interest of 7%.



$$P.V = 1000 + 1000(1.04) + \dots + 1000(1.04)^{19}$$

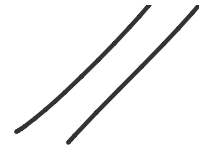
$$= 1000v \left[1 + \frac{(1.04)}{1.07} + \left(\frac{1.04}{1.07} \right)^2 + \dots + \left(\frac{1.04}{1.07} \right)^{19} \right]$$

$$r = \frac{1.04}{1.07}$$

$$= 1000 \cdot \frac{1}{1.07} \cdot \frac{1 - \left(\frac{1.04}{1.07} \right)^{20}}{1 - \left(\frac{1.04}{1.07} \right)}$$

1.07

$$1 - \left(\frac{1.04}{1.07} \right)$$



The present value of this annuity is

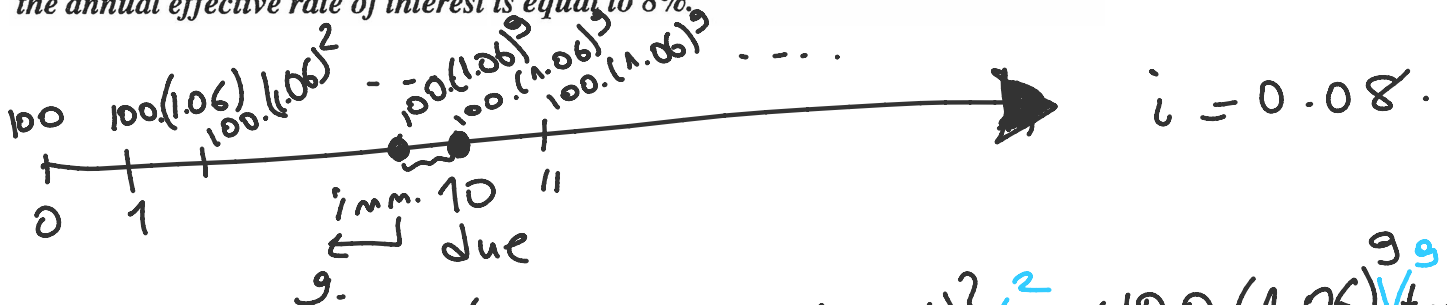
$$1000 \left[\frac{1}{1.07} + \frac{1.04}{(1.07)^2} + \dots + \frac{(1.04)^{19}}{(1.07)^{20}} \right]$$

which can be evaluated using formula (4.29) to give

$$1000 \frac{1 - \left(\frac{1.04}{1.07} \right)^{20}}{.07 - .04} = \$14,459$$

to the nearest dollar.

Example 4.13 A perpetuity-due makes annual payments which begin at \$100 for the first year, then increase at 6% per year through the 10th year, and then remain level thereafter. Calculate the present value of this perpetuity, if the annual effective rate of interest is equal to 8%.



$$P.V = 100 + 100 \cdot (1.06) + 100 \cdot (1.06)^2 + \dots + 100 \cdot (1.06)^9 + \dots$$

$$= 100 \left[1 + \frac{1.06}{1.08} + \left(\frac{1.06}{1.08} \right)^2 + \dots + \left(\frac{1.06}{1.08} \right)^9 \right] + \text{Remain.}$$

$$= 100 \cdot \frac{1 - \left(\frac{1.06}{1.08} \right)^{10}}{1 - \frac{1.06}{1.08}}$$

The present value of the first 10 payments is $100 \left[1 + \frac{1.06}{1.08} + \dots + \left(\frac{1.06}{1.08} \right)^9 \right]$

$$= 100 \left[\frac{1 - \left(\frac{1.06}{1.08} \right)^{10}}{1 - \frac{1.06}{1.08}} \right] = 920.65.$$

Remaining = $100 \cdot (1.06)^9 \cdot \frac{1}{0.08} \cdot v^9$

$\frac{1}{i}$ (per-ann.)

$v^9 \rightarrow v = (1.08)^{-1}$

The present value of the rest of the payments is $100(1.06)^9 \left[\frac{1}{(1.08)^{10}} + \frac{1}{(1.08)^{11}} + \dots \right]$

$$= 100 \left(\frac{1.06}{1.08} \right)^9 \left[\frac{1}{1.08} + \frac{1}{(1.08)^2} + \dots \right] = 100 \left(\frac{1.06}{1.08} \right)^9 \frac{1}{.08} = 1056.45.$$
